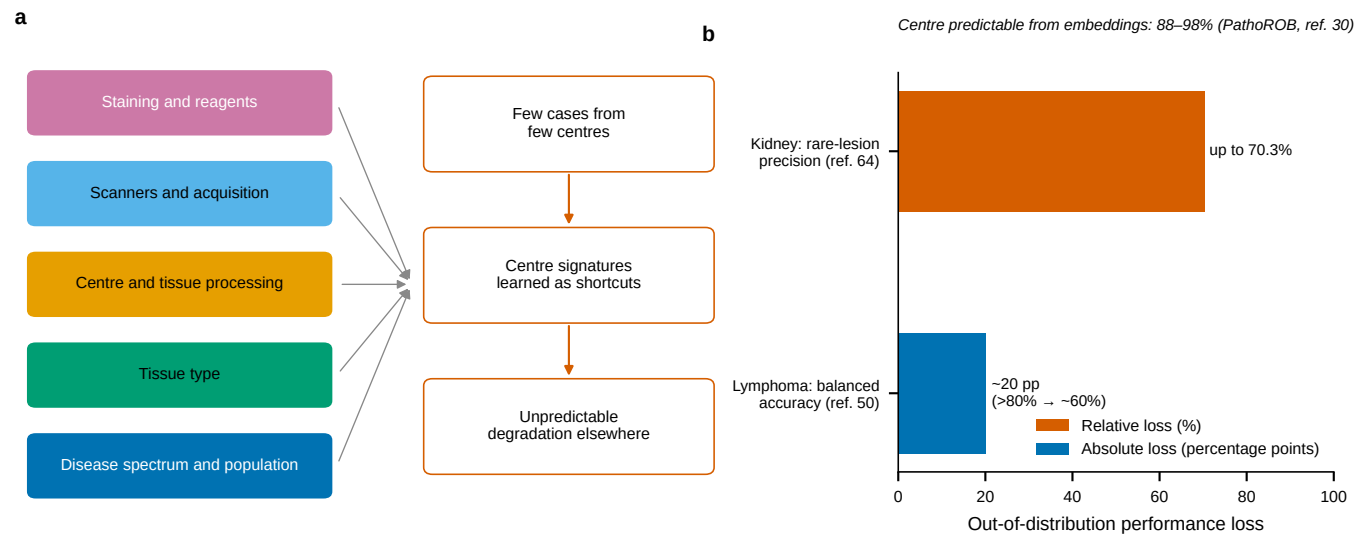
a
Staining and reagents
Scanners and acquisition
Centre and tissue processing
Tissue type
Disease spectrum and population
Few cases from few centres
Centre signatures learned as shortcuts
Unpredictable degradation elsewhere
b
Centre predictable from embeddings: 88–98% (PathoROB, ref. 30)
Kidney: rare-lesion precision (ref. 64)
up to 70.3%
Lymphoma: balanced accuracy (ref. 50)
~20 pp (>80% → ~60%)
Relative loss (%)
Absolute loss (percentage points)
0
20
40
60
80
100
Out-of-distribution performance loss